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Master of Computer Applications

Semester – III

Data Analytics for Business

Experiment No.: 09

Title: Exploratory Data Analysis on Business Datasets Dashboard

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Title: Exploratory Data Analysis on Business Dataset

1. Problem Statement

You are a Data Analyst at a consulting firm. A client has provided a large business dataset and wants to understand key trends and patterns. The objective is to perform exploratory data analysis (EDA) to identify meaningful business insights, understand sales and profit performance, compare customer and product segments, and identify opportunities for improvement using an interactive E-commerce Sales Dashboard.

2. Methodology

Step 1: Data Collection

- Collected an e-commerce business dataset containing sales transaction details.
- The dashboard includes measures such as Amount, Quantity, Profit and Average Order Value (AOV).
- The data also contains State, Category, Sub-Category, Customer Name, Payment Mode and Month.
- The dataset was selected to understand sales distribution, customer contribution, payment preferences and profitability.

Step 2: Data Import

- Opened the business intelligence dashboarding environment.
- Imported the e-commerce sales dataset.
- Loaded the required fields and verified that the data was available for analysis.

Step 3: Data Cleaning and Transformation

- Checked the dataset for missing and null values.
- Checked for duplicate records.
- Verified data types for numerical and categorical columns.
- Ensured Amount, Quantity, Profit and AOV were treated as numerical measures.
- Ensured State, Category, Sub-Category, Customer Name and Payment Mode were correctly formatted.

Step 4: Dashboard Creation

- Created KPI cards to summarize major business measures.
- Created charts for state-wise sales, category-wise quantity, monthly profit, customer sales, payment mode and sub-category profit.
- Added Quarter and State filters for interactive analysis.

3. KPI Cards Analysis

1. Total Amount

- The dashboard shows approximately 438K as the total sales amount.
- It represents the overall sales value generated from the transactions.
- This KPI helps management understand the scale of business revenue.

2. Total Quantity

- The dashboard shows a total quantity of approximately 5,615 units.
- It represents the total number of products sold.
- This helps measure sales volume and product movement.

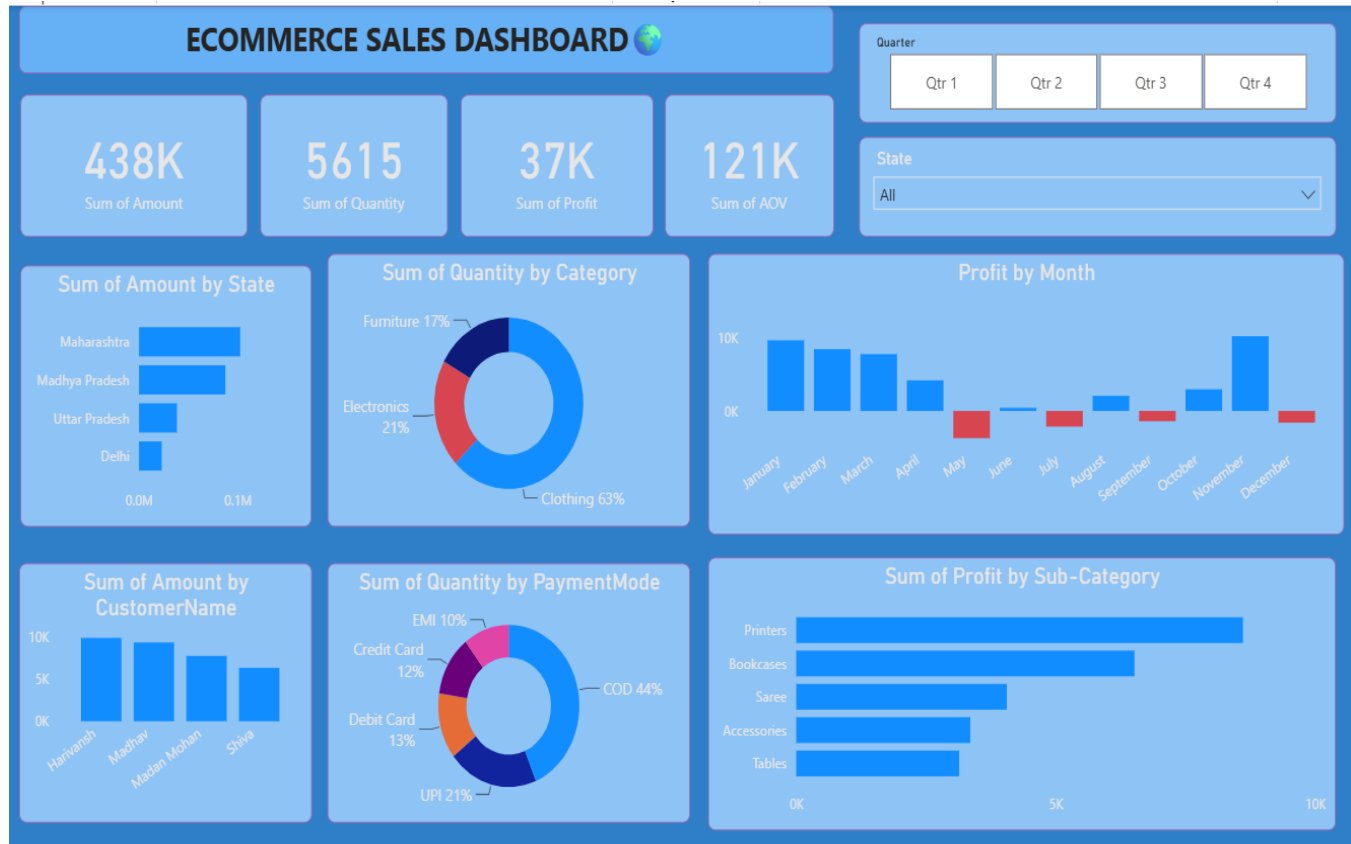
3. Total Profit

- The dashboard shows approximately 37K in total profit.
- It indicates the overall profit generated from the business transactions.
- Management can use this KPI to evaluate the financial performance of sales.

4. Sum of AOV

- The dashboard displays approximately 121K for the AOV measure.
- AOV is used to understand the value associated with customer orders.
- This KPI can help management evaluate order-value patterns and customer spending.

4. Visualization Analysis



1. Amount by State

Purpose: Compares sales amount across different states.

Observation:

- Maharashtra has the highest displayed sales amount, followed by Madhya Pradesh.
- Uttar Pradesh and Delhi contribute lower amounts compared with the top two displayed states.
- The chart indicates that sales are concentrated more strongly in a few states.

Business Insight:

- Maharashtra appears to be an important market and can be studied for successful sales practices.
- Lower-performing states may require targeted marketing, product availability or customer acquisition strategies.

2. Quantity by Category

Purpose: Shows the contribution of each product category to total quantity sold.

Observation:

- Clothing contributes the largest share at approximately 63%.
- Electronics contributes approximately 21%.
- Furniture contributes approximately 17%.

Business Insight:

- Clothing is the strongest category by sales quantity and should receive attention for inventory planning.
- Electronics and Furniture provide additional opportunities for category-specific promotions and cross-selling.

3. Profit by Month

Purpose: Examines how profit changes across the months.

Observation:

- Profit is positive in several months, with November showing the highest visible profit at around 10K.
- January, February and March also show relatively strong positive profit.
- May, July, September and December show negative profit in the dashboard.
- June is close to break-even, while August and October show moderate positive profit.

Business Insight:

- The monthly variation indicates that profitability is not uniform throughout the year.
- Management should investigate the causes of negative-profit months, such as discounts, product mix, shipping costs or returns.

4. Amount by Customer Name

Purpose: Compares the sales amount generated by major displayed customers.

Observation:

- Harivansh has the highest displayed amount at approximately 10K.
- Madhav follows with approximately 9.5K, while Nandan Mohan and Shiva contribute lower amounts.
- The chart shows that a small group of customers contributes a noticeable share of sales.

Business Insight:

- High-value customers can be targeted with loyalty programs and personalized offers.
- Understanding repeat purchases from these customers can help improve customer retention.

5. Quantity by Payment Mode

Purpose: Shows the distribution of sold quantity across payment methods.

Observation:

- COD is the largest payment mode at approximately 44%.
- UPI contributes around 21%, followed by Debit Card at 13%, Credit Card at 12% and EMI at 10%.
- Cash on Delivery remains the dominant payment preference in the displayed data.

Business Insight:

- The business should continue supporting COD while encouraging digital payments through convenient checkout options or suitable incentives.
- Payment preferences can also be analyzed by state and customer segment for more targeted strategies.

6. Profit by Sub-Category

Purpose: Compares profit contribution across selected product sub-categories.

Observation:

- Printers show the highest displayed profit at approximately 8.5K.
- Bookcases contribute around 6.1K, followed by Saree, Accessories and Tables.
- The chart shows clear differences in profitability among sub-categories.

Business Insight:

- High-profit sub-categories such as Printers and Bookcases can be prioritized for inventory and promotional planning.
- Lower-profit sub-categories should be reviewed for pricing, discounts and cost efficiency.

5. Interactive Filters

- Quarter – allows users to compare performance across Qtr 1, Qtr 2, Qtr 3 and Qtr 4.
- State – allows the dashboard to be filtered for a specific state or all states.
- These filters make it easier to analyze business performance for selected periods and geographical markets.

6. Overall Insights

- The dashboard reports approximately 438K in total sales amount.
- A total quantity of approximately 5,615 units is displayed.
- Overall profit is approximately 37K.
- The dashboard displays approximately 121K for the AOV measure.
- Maharashtra has the highest displayed sales amount among the states shown.
- Clothing is the dominant category by quantity with approximately 63% of the displayed share.
- COD is the most used payment mode, contributing approximately 44% of quantity.
- November shows the highest visible monthly profit, while several months show negative profit.
- Printers are the strongest displayed sub-category in terms of profit.
- A small number of customers, particularly Harivansh and Madhav, contribute comparatively high sales amounts.
- The dashboard combines sales, quantity, customer, payment and profitability views to provide a broad understanding of business performance.
- Quarter and State filters support interactive analysis and allow management to investigate specific business segments.

7. Conclusion

The E-commerce Sales Dashboard provides an interactive and simple way to perform exploratory data analysis on a business dataset. It summarizes important measures such as total amount, quantity, profit and AOV through KPI cards and visualizations. The dashboard helps identify state-wise sales patterns, category-wise product demand, monthly profit trends, major customers, payment preferences and profitable sub-categories. The analysis shows that Clothing contributes the largest quantity share, COD is the most common payment method, Maharashtra has the highest displayed sales amount, and Printers are among the strongest profit-generating sub-categories. Monthly profit also varies considerably, indicating opportunities to investigate and improve low-profit periods. Overall, the dashboard helps management understand business trends, identify high-performing segments, detect areas for improvement, and make better data-driven decisions regarding sales, inventory, customers, payments and profitability.

<https://github.com/lohithslohiths110405-ui/Data-Analytics-for-Business/tree/main/EDA%20Dashboard>